Hugo Marquerie 02/05/2025

Teorema de Cauchy-Goursat para discos

Teorema 1 (de Cauchy-Goursat para discos). *Sea* $f \in \mathcal{H}(D(z_0, r))$

$$\implies \forall \gamma \text{ camino cerrado en } D(z_0, r) : \int_\gamma f(z)\,\mathrm{d}z = 0.$$

***Demostración*:** Como $f \in \mathcal{H}(D(z_0, r))$, el teorema de existencia de primitiva nos indican que $\exists F \in \mathcal{H}(D(z_0, r))$ tal que $F' = f$.

$$\implies \int_\gamma f(z)\,\mathrm{d}z = \int_\gamma F'(z)\,\mathrm{d}z \overset{\substack{\text{RB}\\ \downarrow}}{=} F(\gamma(b)) - F(\gamma(a)) = 0 \implies \int_\gamma f(z)\,\mathrm{d}z = 0.$$

■

Referenciado en

- `teo-formula-integral-cauchy-disco`

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